SEMAPHORES

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Write a C program to implement the **Producer–Consumer problem** using semaphores. The program should simulate a bounded buffer where a producer produces items and a consumer consumes items.

The program must use the following variables:

- `mutex` → to ensure mutual exclusion
- `full` → to count filled slots
- `empty` → to count empty slots
- `x` → to track number of items

The user should be able to choose:

1. Producer
2. Consumer
3. Exit

Sample Input 1:

1

1

2

3

Sample Output 1:

----- Producer Consumer Problem -----

1. **Producer**
2. **Consumer**
3. **Exit**

Enter your choice: 1 Producer produces item 1

Enter your choice: 1 Producer produces item 2

Enter your choice: 2 Consumer consumes item 2

Enter your choice: 3 Exiting program... Final buffer items produced: 1

Sample Input 2:

2

1

3

Sample Output 2:

----- Producer Consumer Problem -----

Enter your choice: 2 Buffer is empty!! Consumer cannot consume.

Enter your choice: 1 Producer produces item 1

Enter your choice: 3 Exiting program... Final buffer items produced: 1

CODE:

```
// Producer Consumer


#include <stdio.h>  #include <stdlib.h>

int mutex = 1, full = 0, empty = 5, x =

0;  int wait(int s)

{     return -

-s;  }
```

```
int signal(int s) {
return ++s;
}
void producer()
{
   mutex = wait(mutex);      empty =
wait(empty);      full = signal(full);      x++;
printf("\nProducer produces item %d", x);
mutex = signal(mutex);
}

void consumer()
{
   mutex = wait(mutex);


   full = wait(full);
empty = signal(empty);
printf("\nConsumer
consumes item %d", x);

   x--;
```

```
    mutex = signal(mutex);
}


int main()
{    int n;


    printf("\n----- Producer Consumer Problem -----");
printf("\n1. Producer");     printf("\n2. Consumer");     printf("\n3.
Exit");

    while(1)
    {
        printf("\nEnter your choice: ");
scanf("%d",&n);


        switch(n)
        {
case 1:
if(mutex==1
&&
empty!=0)
```

```
            {
producer();
            }
else
            {
               printf("\nBuffer is full!! Producer cannot produce.");
            }
break;

         case 2:
            if(mutex==1 && full!=0)
            {
consumer();
            }
else
            {
               printf("\nBuffer is empty!! Consumer cannot consume.");
            }
break;
case 3:
```

```
                printf("\nExiting program...");
printf("\nFinal buffer items produced: %d\n", x);
                exit(0);

            default:
                printf("\nInvalid choice. Try again.");
        }
    }

    return 0;
}
```